Electric Vehicle Market Performance & Insights

Project Documentation | Tableau Data Analytics Project Prepared by: Janhvi Sisodiya | Platform: SkillWallet | Date: 24 October 2026

1. Project Overview

The electric vehicle (EV) industry is one of the fastest-growing sectors of the global automotive market. This project analyses an EV market dataset to answer questions asked by three business personas:

- **Emily Johnson (Market Strategy Director):** Which vehicle categories drive annual sales, who are the leading manufacturers, and how satisfied are customers?
- **David Chen (Business Intelligence Manager):** What are the manufacturer market shares, battery technology distribution, and manufacturing country trends?
- **Dr. Sarah Williams (Sustainable Mobility Research Analyst):** How do battery technologies, driving range, and manufacturing distribution compare across EV categories?

The result is an interactive Tableau dashboard, a data story with recommendations, and a Flask web application that embeds both.

2. Tools and Technologies

Tool	Purpose
Tableau Public (Desktop Public Edition)	Data connection, charts, dashboards, story
Tableau Public (web)	Publishing and sharing
Python 3 and Flask	Web integration (embedding dashboards)
HTML / CSS / JavaScript	Flask page layout and tabs
Kaggle	Dataset source

3. Dataset

- **Name:** Electric Vehicle (EV) Market Classification Dataset (Kaggle, author: Harish Reddy)
- **File used:** ev_market_train.csv
- **Size:** 200,000 rows and 37 columns
- **Key columns:** manufacturer, model_year, manufacturing_country, market_region, category, battery_type, battery_capacity_kwh, range_km, price_usd, annual_sales_units, customer_rating, safety_rating_stars, recommended_use_case
- **Coverage:** 200 manufacturers, 15 vehicle categories, 5 battery types, 10 manufacturing countries, 7 market regions, model years 2018 to 2026

4. Data Preparation

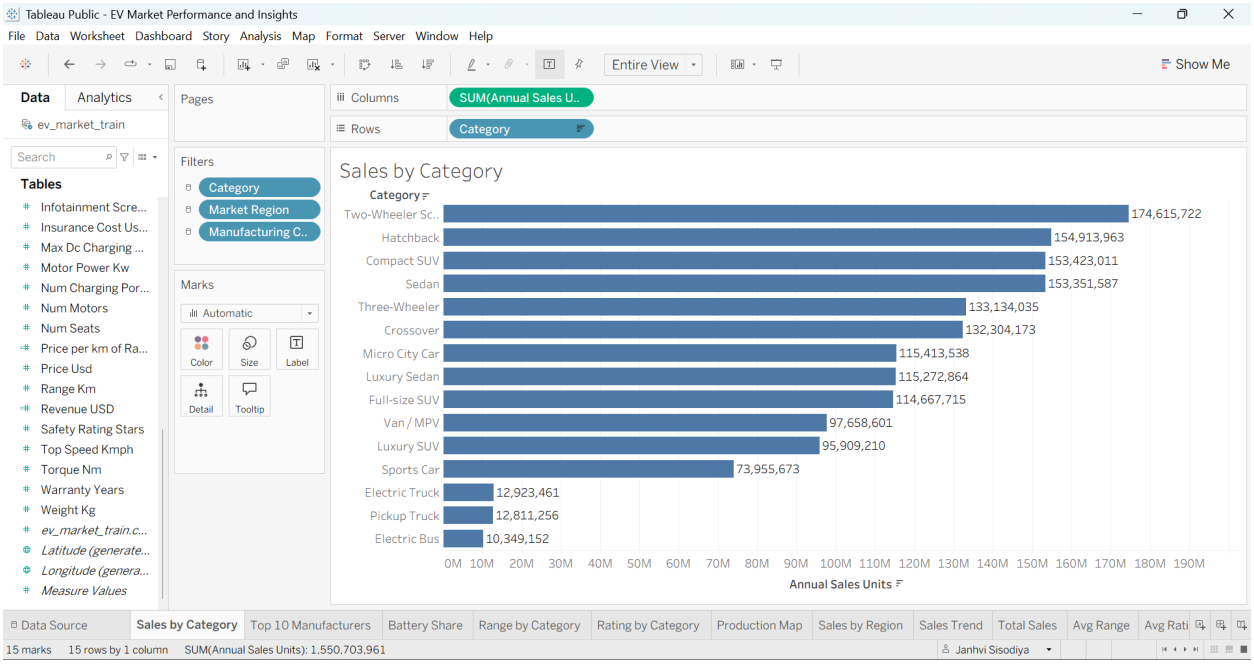
1. Opened Tableau Public and connected the CSV through **Connect > To a File > Text file**. (The Microsoft Excel connector rejects CSV files with an "Illegal file type for Excel protocol" error.)
2. Checked data types on the Data Source page: text columns as strings, numeric columns as numbers.
3. Reviewed missing values. Blanks exist in annual_sales_units (2,028 rows), customer_rating (3,977), resale_value_pct_3yr (6,103), battery_cycle_life (3,786) and infotainment_screen_inch (3,102). Tableau ignores blanks in sums and averages, so no rows were deleted.
4. Set the geographic role of manufacturing_country to **Country/Region** so it can be plotted on a map.
5. Rule followed in all charts: **SUM** for annual_sales_units, **AVG** for range_km, customer_rating and price_usd.

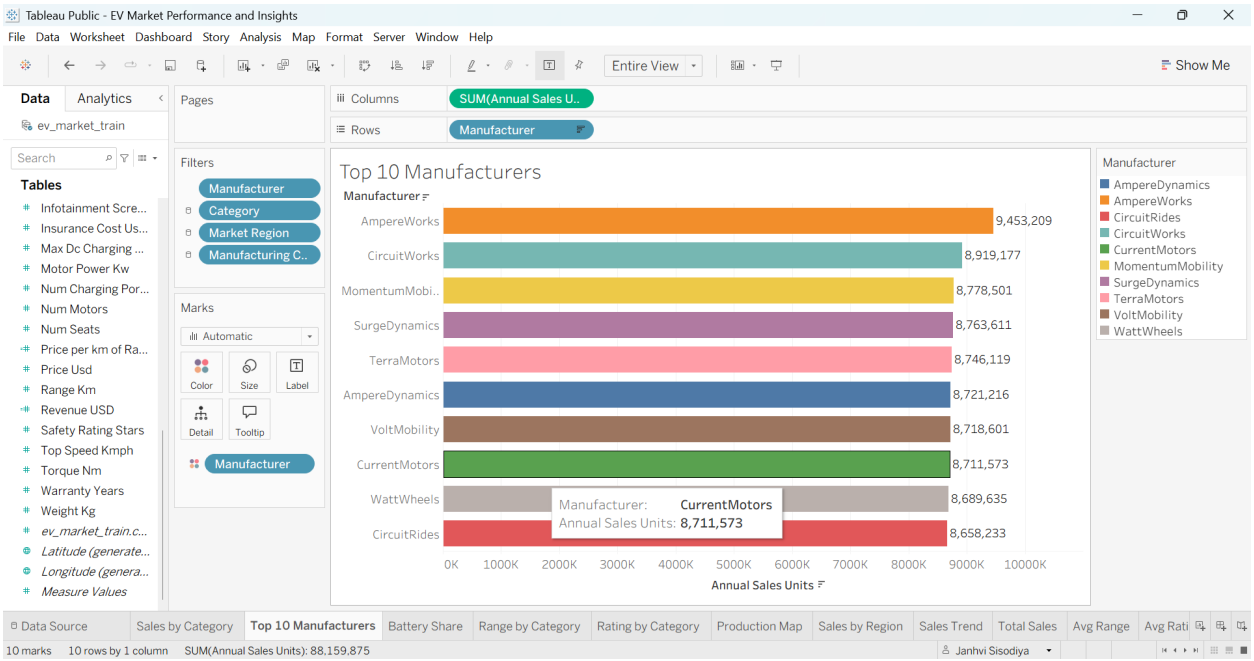
5. Visualizations

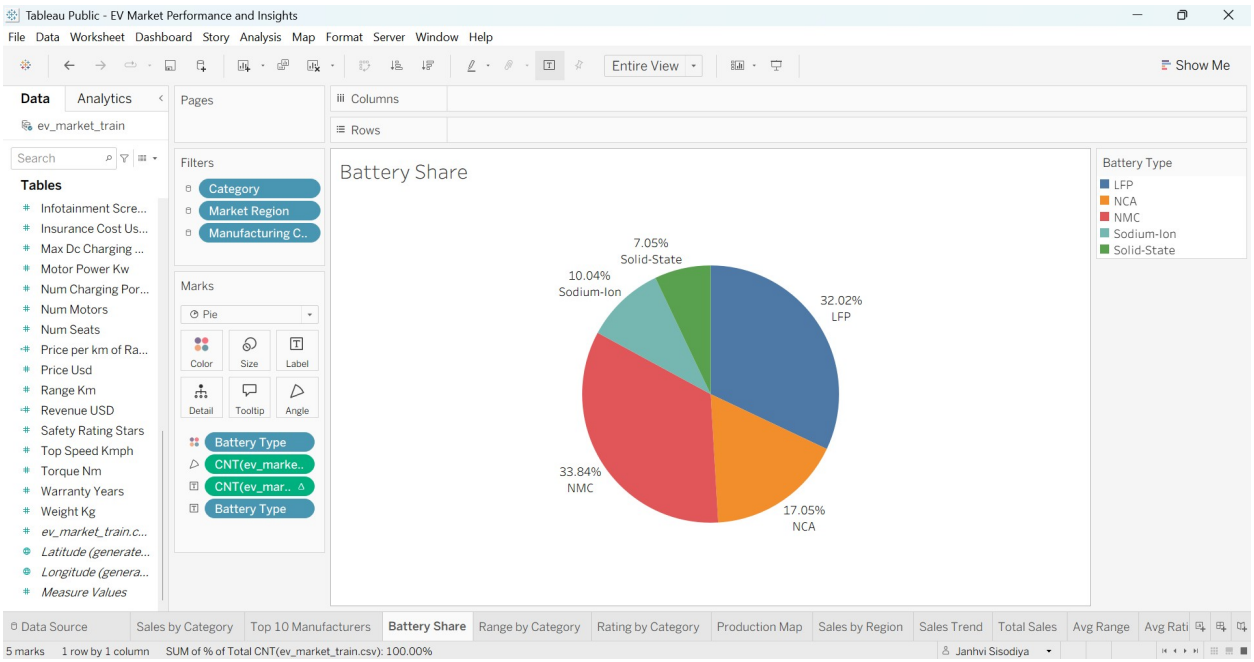
The workbook contains 16 worksheets.

#	Worksheet	Chart type	Business question
1	Sales by Category	Horizontal bar	Which categories contribute most to sales?
2	Top 10 Manufacturers	Horizontal bar (Top 10 filter)	Who are the leading manufacturers?
3	Battery Share	Pie	Which battery technologies dominate?
4	Donut Chart	Donut (dual axis)	Sales share by market region
5	Range by Category	Horizontal bar	How does driving range differ by category?
6	Rating by Category	Horizontal bar	How satisfied are customers by category?
7	Production Map	Map	Where are vehicles manufactured?
8	Sales by Region	Horizontal bar	Which market regions sell the most?
9	Sales Trend	Line	How have sales changed from 2018 to 2026?
10	Heatmap	Heatmap	Category vs region sales intensity
11	Total Sales	KPI card	Headline total sales
12	Avg Range	KPI card	Headline average range
13	Avg Rating	KPI card	Headline average customer rating
14	Revenue USD	Horizontal bar	Which categories generate the most revenue?
15	Price per km of	Horizontal bar	What does each km of range cost by category?

#	Worksheet	Chart type	Business question
	Range		
16	Rating Level Distribution	Horizontal bar	How are vehicles spread across rating levels?







6. Calculated Fields

Three calculated fields were created through **Analysis > Create Calculated Field**.

Field	Formula	Purpose
Revenue USD	`[Annual Sales Units] * [Price Usd]`	Estimated revenue for each vehicle model
Price per km of Range	`SUM([Price Usd]) / SUM([Range Km])`	Cost of one km of driving range
Rating Level	`IF ISNULL([Customer Rating]) THEN "Not Rated" ELSEIF [Customer Rating] >= 4.5 THEN "Excellent" ELSEIF [Customer Rating] >= 4 THEN "Good" ELSE "Average" END`	Groups ratings into levels

7. Dashboards, Filters and Story

7.1 EV Market Dashboard

Top row: three KPI cards. Below: sales by category, top 10 manufacturers, battery share, donut chart, sales trend, range and rating by category, production map, heatmap and sales by region. A text box with key insights sits at the bottom.

7.2 Advanced Analysis Dashboard

Contains the three calculated-field charts (Revenue USD, Price per km of Range, Rating Level Distribution) and a Market Region filter.

7.3 Filters

Filter	Applied to
Market Region	All worksheets using the data source
Category	All worksheets using the data source
Manufacturing Country	All worksheets using the data source
Manufacturer (Top 10 by sales)	Top 10 Manufacturers sheet

Each filter is set to **Apply to Worksheets > All Using This Data Source**, so one selection updates every chart.

7.4 Data Story

The story "EV Market: From Data to Decisions" has six slides: market overview, category sales, manufacturer competition, battery technology, regional spread, and recommendations.

8. Key Insights

All figures below come from the full dataset (200,000 rows).

1. **Total sales** are about 1.55 billion units. Average driving range is 324.5 km and average customer rating is 4.09 out of 5.
2. **Two-Wheeler Scooters lead in units** (about 174.6M), followed by Hatchbacks (about 154.9M), Compact SUVs (about 153.4M) and Sedans (about 153.4M). Electric Buses (about 10.3M), Pickup Trucks (about 12.8M) and Electric Trucks (about 12.9M) sell the least.
3. **AmpereWorks is the top manufacturer** (about 9.45M units), followed by CircuitWorks (about 8.92M) and MomentumMobility (about 8.78M). The gap within the top 10 is small, so competition is intense.
4. **NMC (33.8%) and LFP (32.0%) dominate battery technology.** NCA holds 17.0%, Sodium-Ion 10.0% and Solid-State 7.1%. Average range is almost identical across battery types (about 324 to 326 km), so battery type alone does not explain range.
5. **Range depends on category.** Luxury Sedans (about 551 km), Luxury SUVs (about 525 km) and Full-size SUVs (about 475 km) go farthest. Two-Wheeler Scooters (about 105 km), Three-Wheelers (about 130 km) and Micro City Cars (about 150 km) have the shortest range.
6. **Revenue ranking differs from unit ranking.** Sports Cars (about 12.1 trillion USD estimated), Luxury SUVs (about 11.1 trillion) and Luxury Sedans (about 10.4 trillion) lead in revenue, while scooters lead only in units. Revenue here is units multiplied by price, so it is an estimate based on this dataset.

7. **Electric Buses and Electric Trucks cost the most per km of range** (about 1,018 and 744 USD per km), followed by Sports Cars (about 436). Two-Wheeler Scooters cost the least (about 24).
8. **Customer satisfaction is similar across categories**: average ratings sit between about 4.08 and 4.10. By Rating Level, 41.4% of vehicles are Average (below 4.0), 33.4% Good (4.0 to 4.49), 23.1% Excellent (4.5 and above) and 2.0% Not Rated.
9. **Sales and production are spread evenly.** South Asia (about 224.0M units) and Europe (about 217.5M) are the highest and lowest regions, a gap of under 3%. Among manufacturing countries, India (about 159.2M) leads and Japan (about 148.3M) is lowest.
10. **Yearly sales are stable** between about 166M and 177M units from 2018 to 2026, with no strong growth trend.

9. Recommendations

1. Prioritise high-volume segments (scooters, hatchbacks, compact SUVs, sedans) for volume growth.
2. Treat luxury and sports segments as high-revenue, lower-volume opportunities.
3. Treat buses, trucks and pickups as niche segments and validate demand before heavy investment.
4. Focus on NMC and LFP batteries today and monitor Solid-State and Sodium-Ion as emerging options.
5. Compete on range, price and rating rather than battery type alone.
6. Expand across all regions, since demand is evenly distributed and no single market dominates.

10. Performance Testing

Parameter	Result
Amount of data rendered	1 CSV source, 200,000 rows, 37 columns
Utilisation of filters	4 filters (Market Region, Category, Manufacturing Country, Top 10 Manufacturers)
Number of calculated fields	3
Number of visualizations	16 worksheets (13 charts and map types plus 3 KPI cards), 2 dashboards, 1 story

Optimisation notes: only the needed fields are used in each sheet, Top 10 filtering limits the number of manufacturer marks drawn, and filters are applied through one shared data source.

11. Web Integration

11.1 Publishing to Tableau Public

The workbook was published through **File > Save to Tableau Public**.

- Dashboard:
  https://public.tableau.com/app/profile/janhvi.sisodiya/viz/ElectricvehicleMarketPerformanceandInsights/EVMarketDashboard
- Advanced Analysis:
  https://public.tableau.com/app/profile/janhvi.sisodiya/viz/ElectricvehicleMarketPerformanceandInsights/AdvancedAnalysis
- Story:
  https://public.tableau.com/app/profile/janhvi.sisodiya/viz/ElectricvehicleMarketPerformanceandInsights/Story1

11.2 Embedding with Flask

A small Flask application serves one page with three tabs (Dashboard, Advanced Analysis, Story). Each tab uses the Tableau Embedding API v3 web component.

```
ev_flask_app/
  app.py
  requirements.txt
  templates/
    index.html
```

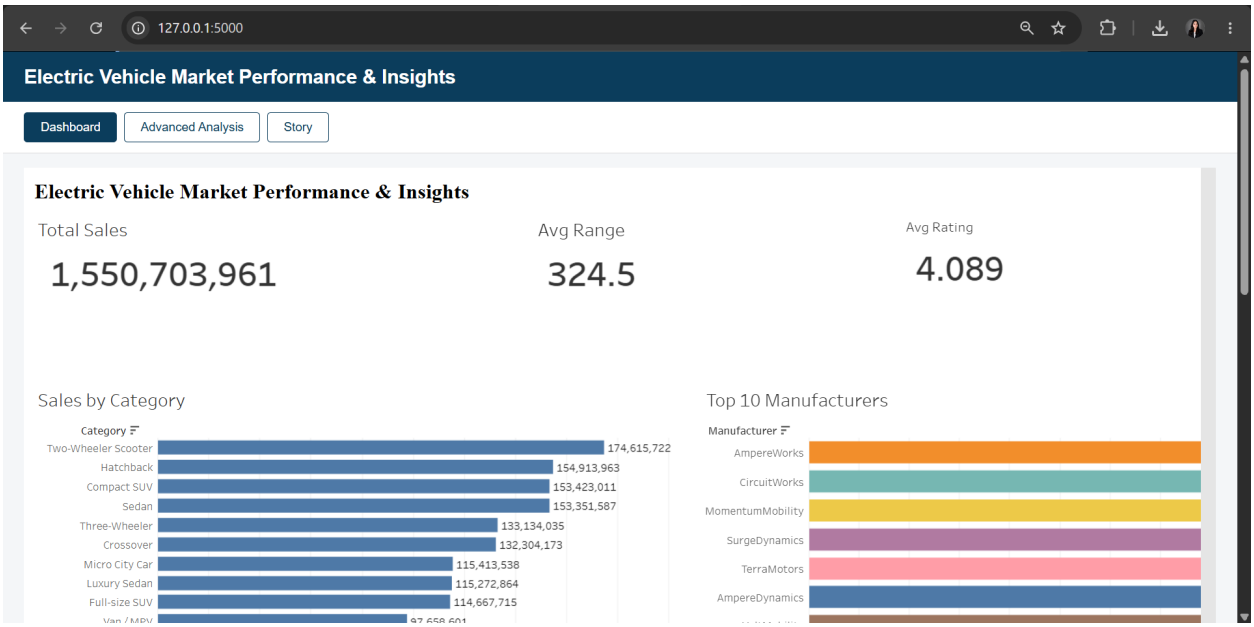
- `app.py` defines the three Tableau URLs and renders `index.html` at the route `/`.
- `index.html` loads the Tableau embedding script and shows each view in a `tableau-viz` element, with simple JavaScript tab switching.

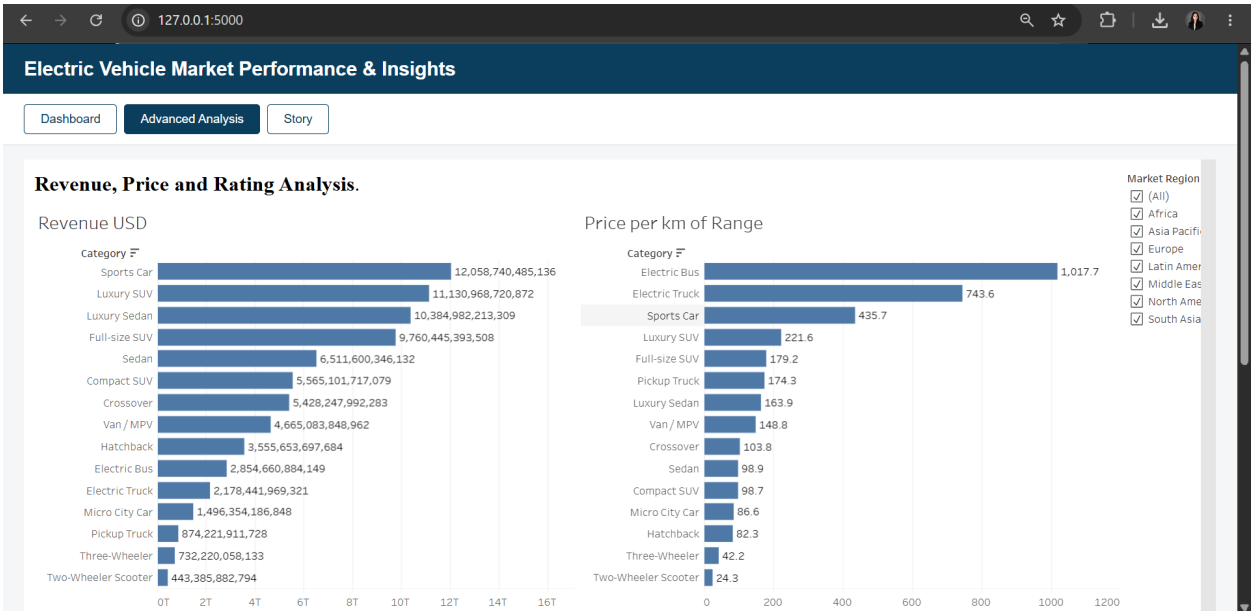
Run locally:

```
pip install -r requirements.txt
python app.py
```

Then open http://127.0.0.1:5000 in a browser.

Troubleshooting note: Flask only reads templates from a folder named `templates`. Placing `index.html` elsewhere gives the error `TemplateNotFound: index.html`.





12. Step-by-Step Development Procedure

1. Understood the three business scenarios and listed the required insights.
2. Downloaded the dataset from Kaggle.
3. Installed Tableau Public and connected ev_market_train.csv using the Text file connector.
4. Checked data types, missing values and geographic roles.
5. Built the sales, manufacturer, battery, range, rating, region, trend, map and heatmap worksheets.

6. Built three KPI worksheets (Total Sales, Avg Range, Avg Rating).
7. Created three calculated fields and three worksheets that use them.
8. Assembled the EV Market Dashboard and the Advanced Analysis dashboard.
9. Added filters and applied them to all worksheets.
10. Wrote the key insights text box.
11. Built the six-slide data story with recommendations.
12. Published the workbook to Tableau Public.
13. Built the Flask application and embedded the dashboards and story.
14. Ran performance checks and recorded the results.
15. Recorded the explanation video and prepared this documentation.

13. Limitations

- The dataset appears to be synthetic: several measures (range by battery type, sales by country and region) are almost uniform. Conclusions describe this dataset, not the real EV market.
- Revenue is estimated as units multiplied by price and is not a reported financial figure.
- Some rows have blank sales or ratings and are excluded from those calculations.

14. Conclusion

The project turns a 200,000-row EV dataset into an interactive dashboard, a decision-focused story and a web-embedded application. It shows that volume is driven by affordable, short-range vehicles, revenue by luxury and sports categories, and that NMC and LFP batteries dominate. Together these insights support product portfolio planning, market expansion and investment decisions.